

One of the most important link for students is Student Zone which can be reached from Home page by selecting Student Zone option on the Student Support Option List (Link address: <http://www.ignou.ac.in/ignou/studentzone>). Figure 3 displays the options of the Student Zone page.

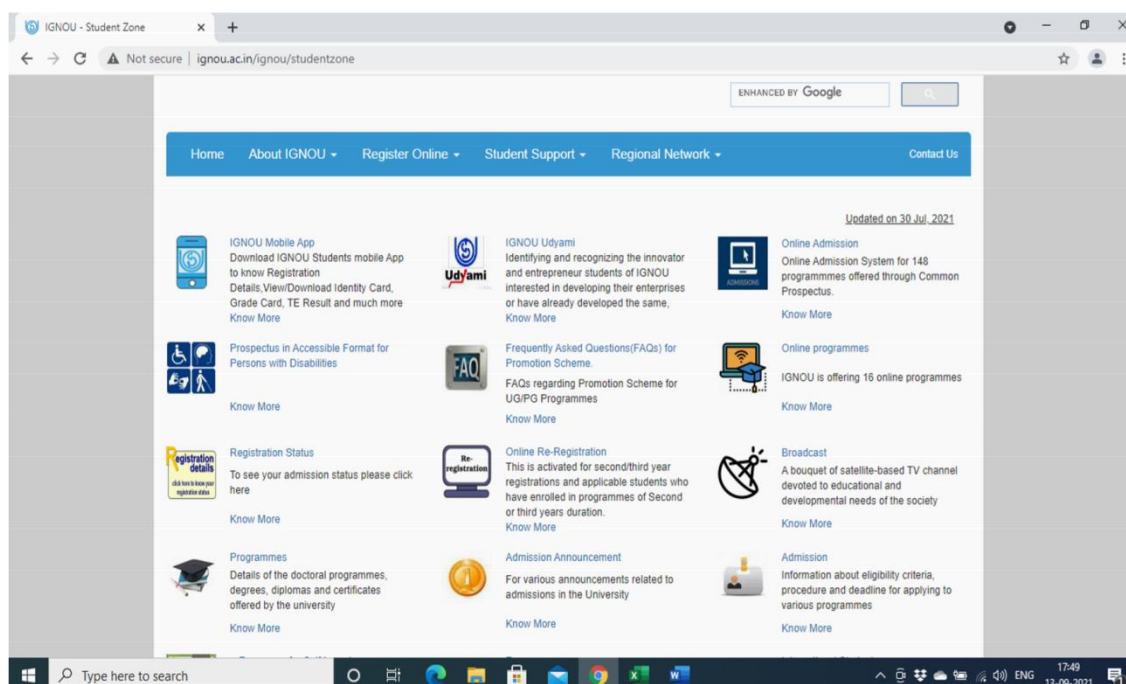


Figure 3: Student Zone page

4. MCA SYLLABUS

SEMESTER - I

MCS-211 Design and Analysis of Algorithms

Credit: 4

Algorithms are the central part of computing and Design and Analysis of algorithms course is the core of the study of Computer Science discipline. The revised course on design and analysis of algorithm introduces many new topics: Deterministic and Stochastic Algorithms , how to solve recurrence relation problems through Substitution method, Recurrence tree and Master methods, An overview of local and global optima ,Fractional Knapsack problem ,Huffman Codes ,a task scheduling algorithm , Topological Sort ,Strongly Connected Components , Maximum Bipartite Matching Problem, Binomial coefficient computation , Floyd Warshall algorithm , **String Matching Techniques** :The naïve String Matching Algorithm, The Rabin Karp Algorithm, Knuth –Morris Pratt Algorithm, **Handling Intractability**: Approximation algorithms for Vertex Cover problem and Minimizing makespan as parallel machines (Graham's algorithm) , Parameterized algorithm for Vertex Cover problem and Meta-heuristic Algorithms

Course Structure*

Block- 1 Introduction to Algorithms

Further Readings

Unit 1: Basics of an Algorithm and its properties

- Introduction
- Objective
- Example of an Algorithm
- Basics building blocks of Algorithms
 - A survey of common running time
- Analysis & Complexity of Algorithm
- Types of problems
- Problem Solving Techniques
- Deterministic and Stochastic Algorithms
- Summary
- Solutions/Answers
- Further Readings

Unit 4: Solving Recurrences

- Introduction

Unit 2: Some pre-requisites and Asymptotic Bounds

- Introduction
- Objectives
- Some Useful Mathematical Functions & Notations
 - Functions & Notations
 - Modular Arithmetic/Mod Function
- Mathematical Expectation
- Principle of Mathematical Induction
- Concept of Efficiency of an Algorithm
- Well Known Asymptotic Functions & Notations
- Summary
- Solutions/Answers

Unit 3: Analysis of Simple Algorithm

- Introduction
- Objectives
- complexity Analysis of Algorithms
 - Euclid Algorithm for GCD
 - Polynomial Evaluation Algorithm
 - Exponent Evaluation
 - Sorting Algorithm
 - 3.3 Analysis of Non-Recursive Control Structures Sequencing
 - For Construct
 - While and Repeat Constructs
 - Recursive Constructs
 - Summary
 - Solutions/Answers

- Objective
- Substitution Methods
- Iteration Methods
- Recursive Tree Methods
- Master Methods
- Summary
- Solution/Answers
- Further Readings

Block- 2 Design Techniques-I

Unit 1: Greedy Technique

- Some Examples to understand Greedy Techniques
- Formalization of Greedy Techniques
- An overview of local and global optima
- Fractional Knapsack problem
- Huffman Codes
- A task scheduling algorithm

Unit 2: Divide & Conquer Technique

- General Issues in Divide and Conquer Technique
- Binary Search Algorithm
- Sorting Algorithm ○ Merge Sort ○ Quick Sort
- Matrix Multiplication Algorithm

Unit 3: Graph Algorithm -I

- Basic Definition and terminologies
- Graph Representation ○ Adjacency Matrix ○ Adjacency List
- Graph Traversal Algorithms ○ Depth First Search ○ Breadth First Search
- Topological Sort
- Strongly Connected Components

Block- 3 Design Techniques – II

Unit 1: Graph Algorithms- II

- Minimum Cost Spanning Tree problems
Kruskal's Algorithm

Unit 2: Dynamic Programming Technique

- The Principle of Optimality
- Chained Matrix Multiplication
- Matrix Multiplication Using Dynamic Programming
- Optimal binary search trees problems
- Binomial coefficient computation
- Floyd Warshall algorithm

Unit 3: String Matching Techniques

- The naïve String-Matching Algorithm
- The Rabin Karp Algorithm
- Knuth –Morris Pratt Algorithm
-

Block- 4: NP- Completeness and Approximation Algorithm

Unit-1: NP-Completeness

- Concepts of Class-P, NPCompleteness, NP-Hard , Unsolvable problems
- Polynomial-time
- Polynomial-time Reductions
- Class P with Examples
- Knapsack and TSP problems

Unit 2: NP-Completeness and NP- hard Problems

- Polynomial Time verification
- Techniques to show NP- Hardness
- NP-Complete problems and P Vs NP problems?

Unit 3: Handling Intractability

- Approximation algorithms for Vertex Cover problem and minimizing make- span as parallel machines (Graham's algorithm)
- Parameterized algorithm for Vertex Cover problem
- Meta-heuristic Algorithms

Prim's Algorithm

- Single Source Shortest Path Problems
Bellman Ford Algorithm
Dijkstra's Algorithm
- Maximum Bipartite Matching Problem

Discrete mathematics deal with discrete objects (that is not continuous) like a set of PhD students in computer science departments. Foundation of Computer Science is built upon discrete mathematics. It includes Propositional calculus, sets, relations, function, graphs, Boolean algebra and advanced counting principles. Knowledge of discrete structures helps in analyzing algorithms, and understanding different areas of computer science courses. The revised course on discrete mathematics course includes **Finite State Machines:** Introduction to Finite Automata, Computability and Complexity, Moore and Mealy State Machines and Deterministic Finite Automata, **Regular Expression and Languages:** Formal Definition of Regular expression and language, Building Regular Expressions, Finite Automata and Regular Expressions, Kleene Closure Definition and Algebra of regular Expressions.

- What is a Proof? ○ Properties Common to Some Terminology Logic
 - and Sets ○ Hypothesis • Relations ○ Axioms ○
 - Representing relation using ○ Lemmas matrices
 - Corollary ○ Representing relation using
 - Conjunction digraph
- Different Methods of Proof ○ Cartesian Product ○ Direct Proof ○
 - Relations and their types

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Course Structure***Block-1 Elementary Logic and Proofs**

- Logic Circuits
- Boolean Functions • Summary
- Solutions/Answers

Unit 1: Propositional Calculus

- Introduction • Objectives
- Propositions
- Logical Connectives ○ Disjunction ○ Conjunction ○ Negation ○ Conditional Connectives ○ Precedence Rule
- Logical Equivalence
- Logical Quantifiers
- Application of Propositional Logic ○ Web Page Searching ○ Logic Circuits
- Summary
- Solutions/Answers

Unit 2: Methods of Proof

- Introduction
- Objectives
 - Indirect Proofs ○ Counter examples
- Principle of Mathematical Induction
- Summary
- Solutions/Answers

Unit 3: Boolean Algebra and Circuits

- Introduction
- Objectives
- Boolean Algebras

Block- 2 Sets and Languages**Unit 1: Sets, Relations and Function**

- Introduction
- Objectives
- Introducing Sets
- Operations on Sets ○ Basic Operations
 - Properties of Relations
- Functions ○ Types of Functions ○ Composition of Functions ○ Some Important function ○ Operations on Functions
- Summary
- Solutions /Answers

Unit 2: Finite State Machines

- Introduction to Finite Automata, Computability and Complexity
- Moore and Mealy State Machines
- Deterministic Finite Automata

Unit 3: Regular Expression and Languages

- Formal Definition of Regular expression and language

- Building Regular Expressions
- Finite Automata and Regular Expressions
- Kleene Closure Definition
- Algebra of regular Expressions

Block 3: Counting Principles

Unit 1: Combinatorics

- Introduction
- Objectives
- Multiplication and Addition Principles
- Permutations
 - Permutations of Objects not Necessarily Distinct
 - Circular Permutations
- Combinations
- Binomial Coefficients & Identities
- Summary
- Solutions/Answers

Unit 2: Advanced Counting Principles

- Introduction
- Objectives
- Pigeonhole Principle
- Inclusion-Exclusion Principle
- Applications of Inclusion – Exclusion
 - Application to Subjective Functions
 - Application to Probability
 - Application to Derangements
- Summary
- Solutions/Answers

Unit 3: Recurrence Relations

- Introduction
- Objectives
- Three Recurrent Problems
- Divide and Conquer Technique to solve Recurrence Relation
- Some Other Methods
 - Method of Inspection
 - Method of telescoping Sums
 - Method of Iteration
 - Method of Substitution
- Summary
- Solutions/Answers

Unit 4: Partitions and Distributions

- Introduction
- Objectives
- Integer Partitions
- Distributions
 - Distinguishable Objects

into Distinguishable Containers

- Distinguishable Objects into Indistinguishable Containers
- Indistinguishable Objects into Distinguishable Containers
- Indistinguishable Objects into Indistinguishable Containers

- Summary
- Solutions/Answers

Block-4 Graph Theory

Unit 1: Basic Properties of Graphs

- Introduction
- Objectives
- Graphs
 - Graph Models

- Social

- Trees

MCS-213 Software Engineering (Credits:4)

The objective of the Course is to make the learner efficiently work as software engineer. S/he should be well acquainted with all the phases of Software Development Life Cycle as well as latest topics in Software Engineering. The learner should be able to apply the

Networks ○

Communication Networks ○

Web

Graphs

- Degree, Regularity and Isomorphism
- Subgraphs
- Represent Graphs ○ Adjacency Matrices ○ Adjacency Lists ○ Incidence Matrix
- Summary
- Solutions/Answers

Unit 2: Connectedness

- Introduction
- Objectives
- Connected Graphs ○ Paths, Circuits and Cycles ○ Components ○ Connectivity
- Bipartite Graphs ○ A complete bipartite graph

concepts learned for doing research.

Course Structure*

BLOCK 1: Overview of Software Engineering

Unit1: Software Engineering and its models

- Evolution of Software Engineering
- Software development models
- Capability maturity models
- Software process technology

Unit2: Principles of Software Requirements Analysis

- Engineering the product

- Summary

- Solutions/Answers

Unit 3: Eulerian and Hamiltonian Graphs

- Introduction
- Objectives
- Eulerian Graphs
- Hamiltonian Graphs ○ Dirac's Theorem ○ Ore's Theorem
- Travelling Salesperson Problem
- Summary
- Solutions /Answers

Unit 4: Graph Colouring

- Introduction
- Objectives
- Vertex Colouring
- Edge Colouring
- Planar Graphs
- Map Colouring Problem
- Summary
- Solutions/Answers
- Modeling the system architecture
- Software prototyping and specification

Unit 3 Software Design

- Data design
- Architectural design
- Interface design
- HCI design
- Modular design
- User Experience design
- Design for Mobility
- Pattern based Design

Unit 4 Software Quality and Security

- Quality Concepts
- Quality Assurance
- Security Engineering

BLOCK 2: Software Project Management

Unit 5: Software Project Planning

- Different types of project metrics
- Software project estimation
- Models for estimation
- Automated tools for estimation
- Software Analytics

Unit6: Risk management and Project

Scheduling

- Identification of Software risks
- Monitoring of risks
- Management of risks
- Formulating a task set for the project
- Choosing the tasks of software engineering
- Scheduling methods
- The Software project plan

Unit 7: Software Testing

- Component Level Testing
- Integration Level Testing
- Mobility Testing

Unit 8 Software change management

- Baselines
- Version control
- Change control
- Auditing and reporting

BLOCK 3: Web, Mobile and CASE tools

Unit9: Web Software Engineering

- Different layers

MCS-214 Professional Skills and Ethics (Credits: 2)

- Issues of management of web-based projects

- Metrics
- Analysis
- Design
- Testing

Unit10: Mobile Software Engineering

- Transition from design to coding of mobile applications
- Elements of mobile applications
- Approaches to the development of mobile applications

Unit11: CASE tools

- Analysis tools
- Design tools
- SQA tools
- UI design tools
- Software testing tools
- Web engineering tools

Unit12: Advanced Software Engineering

- Clean room Software engineering
- Component based Software engineering
- Re-engineering
- Reverse engineering

Block-4: Advanced Topics in Software Engineering

- Unit-13: Software Process Improvement
- Unit-14: Emerging Trends
- Unit-15: Introduction to UML • Unit-16: Data Science for Software Engineers

This course is aimed to develop the communicational skills, professional skills and ethics at the work place. In this course, we concentrate on English at the workplace. You are probably wondering whether business English (as it is also called) is a separate language to general English. Certainly not, business English is not a separate language. It is English used at the workplace using specific vocabulary, and in certain situations

having a different discourse. Every profession uses a certain ‘jargon’ and the business context in no different. While business English is firmly rooted in general English, nevertheless there are certain distinguishing features which are evident. In this Course, you will learn some theoretical inputs into the process of communication, its different types, the difference between written and oral communication. We then concentrate on the structure of conversation – its characteristics and conventions, effectively speaking over the telephone, preparing Curriculum Vitae for jobs and interviews, preparing and participating in the Group Discussions, Presentation Skills, Copyright and Plagiarism issues and many more.

- Pronunciation: Contracted Forms

Course Structure*

BLOCK1: Professional Skills Needed at the Work Place - I

Unit 1: The Process of Communication

- Introduction: What is Communication?
- The Process of Communication
- Barriers to Communication • Different Types of Communication
- Written vs. Oral Communication
- Different Types of Face-toFace Interactions
- Characteristics and Conventions of Conversation
- Conversational Problems of Second/Foreign Language Users
- Difference between Conversation and Other Speech Events

Unit2: Telephone Techniques

- Warm Up
- Speaking and Listening: Commonly Used Phrases in Telephone Conversations
- Reading: Conference Calls
- Vocabulary
- Writing and Listening: Leaving a Message
- Grammar and Usage: The Perfect Tenses

Unit3: Job Applications and Interviews

- Warmup
- Reading
- Vocabulary: Apply for a Job
- Curriculum Vitae
- Language Focus: Some Useful Words
- Study Skills: Preparing for an Interview
- Listening
- Speaking
- Writing
- Negotiation Skills

Unit4: Group Discussions

- Reading
- Writing Skills
- Listening: How to be Successful in a Group Discussion
- Study Skills
- Language Focus
- Vocabulary
- Speaking
- Grammar: Connectives
- Pronunciation

Unit5: Managing Organisational Structure

- Warm Up: Ability to Influence and Lead
- Reading: The Role of a Manager
- Vocabulary: Leadership
- Speaking and Listening

- Language Focus: Degree of Probability
- Grammar: Modals
- Writing: Reports
- Pronunciation

BLOCK 2: Professional Skills Needed at the Workplace - II

Unit6: Meetings

- Reading: A Successful Meeting
- Speaking: One to One Meetings
- Language Focus: Opening, Middle and Close
- Study Skills: Editing
- Listening: Criteria for Successful Meetings
- Vocabulary
- Grammar: Reporting Verbs
- Writing: Memos
- Pronunciation: Stress According to Part of Speech

Unit7: Presentation Skills -I

- Reading: Presentation Skills
- Grammar: Verbs often Required in Presentations

Successful Life at the Workplace

- The Changing Scenario in the Twentyfirst Century
- What Employers Want
- Qualities of a Star Performer
- Personal Competence
- Social Competence
- Neurolinguistic Programming (NLP)
- Implementing the Change
- Knowing Who and What Trigger You
- Becoming Aware of Our 'Blind Spots' and Learning to Overcome Them
- Collaboration and Cooperation

Unit10: Work Ethics and Social Media Etiquette

- Ethics at the workplace
- A Talk on Ethics
- Vocabulary: Positive Qualities
- Talking about Ethical and Unethical

MCS-215 Course: Security and Cyber Laws

(Credits:2)

This course introduces the students to some of the latest topics in the context of computer security and cyber laws. These topics are very relevant in the present time.

- Language Focus
- Listening: Importance of Body Language in Presentations
- Speaking: Preparing an Outline of a Presentation
- Pronunciation

Unit8: Presentation Skills –II

- Reading: Structure of Presentation
- Study Skills: Visual Aids
- Ending the Presentation
- Language Focus: Talking about Increase and Decrease
- Grammar: Prepositions
- Listening: Podium Panic
- Speaking
- Pronunciation: Emphasizing the Important Words in Context

Unit9: Developing Interpersonal Skills for a

- Practices
- Improving our Ethics

Unit11: Copyright and Plagiarism

- A Brief History of Copyright
- Evolution of Copyright Law in India • Who Owns a Copyright?
- Economic, Moral and Other Such Rights
- Plagiarism
- What needs to be acknowledged?

Course Structure* Block-1: Cyber Security Issues Unit-1: Cyber security issues and challenges (Will be Adapted from MIR-11 Unit-7, PGCCCL)

- Introduction
- Objectives
- Digital Security: Pros & Cons
- Security Issues /breaches in Cyberspace
- Technology's Answers to Cyber Security
- Cyber Security and the Law

Unit-2: Cryptography Mechanisms (Will be Adapted from MIR-11 Unit-8, PGCCL)

- Introduction
- Objectives
- Introduction to Cryptography
- Functions of Cryptography • Steganography
- Encryption and Decryption
- Encryption Scheme: Public Key and Private Key Distribution
- Commonly used Crypto Algorithms
- RSA and DES
- Electronic Signature
- Authentication and Authorisation
- Hash Functions
- Access Control Derivatives/Mechanisms
- Public Key Infrastructure/ Data Encryption Standard

Unit-3: Data Security and Management (Will be Adapted from MIR-14 Unit-5, PGCCL)

- Introduction
- Objectives
- Security Requirements(CIA)
- Security Threats and Attacks
- Computer, Mobile and Internet
- Security Measures and Solutions
- Security Policy
- Security Management
- Security Audit
- Security & Usability

Block-2: Cyber Laws

Unit-1: Regulation of Cyberspace: An Overview (Will be Adapted from MIR-11 Unit-9, PGCCL)

- Introduction
- Objectives
- Desirability of Regulation of Cyberspace
- Need for Regulation of Cyberspace
- How Cyberspace can be regulated
- Legal and Self-Regulatory Framework
- Filtering devices and Rating Systems
- Government Policies and Laws Regarding Regulation of Internet Content
- UNCITRAL Model Law,1996
- Regulation of Cyberspace Content: Global Scenario

- United States
- European Union
- United Kingdom
- Regulation of Cyberspace Content in India
- International Initiatives for Regulation of Cyberspace
- Organisation for Economic Cooperation and Development(OECD)
- UNESCO
- BRICS

Unit-2: Cyber Crimes

- Introduction
- Objectives
- Classification of Cyber Crimes
- Penalties and compensation (Chapter IX) under IT Act, 2000
- Offences (chapter XI) under IT Act, 2000
- Investigation and procedure ((Chapters XII- 77A to 78 and80)
- Basics of Cyber Forensic
- Cyber Forensic Investigation Tools

- Patents
- Linking, In-lining and framing
- Introduction • Trademark Issues
- Objectives • Domain Name Disputes – Cyber squatting
- Basic Concept: IPRs • Search Engines and their Abuse
- Copyright issues in digital- medium, music • Regulatory Framework- National and International Scenario.

MCSL-216 DAA and Web Design Lab Credits: 2

Main objective of this laboratory course is to provide hands on exercises to the learners based on DAA and Web Design Course.

Lab Sessions:

- There will be 20 practical sessions (3 hours each) of which 10 sessions will be on DAA and 10 sessions will be on Web Designing.
- The practice problems for all 20 sessions will be listed session-wise in the lab manual.

MCSL-217 Software Engineering Lab Credits: 2

Main objective of this laboratory course is to provide hands on exercises to the learners based on Software Engineering Course.

Lab Sessions:

- There will be 20 practical sessions (3 hours each)
- The practice problems for all 20 sessions will be listed session-wise in the lab manual.

Syllabus of Bridge Courses (*Only for the students who are required to register for these courses based on eligibility conditions for admission to MCA programme*)

MCS-201 Programming in C and Python Credits– 4

Objective

The course is aimed to develop problem-solving strategies, techniques and skills that can be applied to computers and problems in other areas which give students an introduction to computer and analytical skills to use in their subsequent course work and professional development. Emphasis of this course is to act as an introduction to the thinking world of computers, to help students develop the logic, ability to solve the problems by using C & Python as programming languages. Knowledge in a programming language is prerequisite to the study of most of computer science courses. This knowledge area consists of those skills and concepts that are essential to problem solving and programming practice independent of the underlying paradigm. The student will learn various concepts and techniques for problem solving and will implement those ideas using C and Python programs.

Block and Unit Structure

C PROGRAMMING (BLOCK - 1 & 2)

BLOCK 1: An Introduction to C

Unit 1: Programming Fundamentals

- Problem - Solving Techniques
- Steps for Problem - Solving

Features of Algorithm

- Flowcharts
- Basic Symbols used in Flowchart Design
- Structured Programming concepts
- C Language and its features
- Structure of a C Program
- Writing a C Program

- Compiling a C Program ○ The C Compiler ○ Syntax and Semantic Errors
- Link and Run the C Program ○ Run the C Program through the Menu ○ Run from an Executable File ○ Linker Errors ○ Logical and Runtime Errors ○

Unit 2: Data Types Operators and Expressions

- Character Set
- Identifiers and Keywords ○ Rules for Forming Identifiers ○ Keywords
- Data Types and Storage
- Data Type Qualifiers
- Variables
- Declaring Variables
- Initialising Variables
- Constants ○ Integer Constants ○ Floating Point Constants ○ Character Constants ○ String Constants ○ Symbolic Constants
- Preprocessor directives
- Assignment Statements
- Arithmetic Operators
- Relational Operators
- Logical Operators
- Comma and Conditional Operators
- Type Cast Operator
- Size of Operator
- C Shorthand
- Priority of Operators

Unit 3: Decision and Loop Control Statements

- Decision Control Statements ○ The *if* Statement ○ The *switch* Statement
- Loop Control Statements ○ The *while* Loop ○ The *do-while* Statement ○ The *for* Loop ○ The Nested Loop
- The *Goto* Statement

- The *Break* Statement
- The *Continue* Statement

Unit 4: Arrays & Strings

- Array Declaration ○ Syntax of Array Declaration ○ Size Specification
- Array Initialization ○ Initialization of Array Elements in the Declaration ○ Character Array Initialization
- Subscript
- Processing the Arrays
- Multi-Dimensional Arrays ○ Multi-Dimensional Array Declaration ○ Initialization of Two-Dimensional Arrays
- Declaration and Initialization of Strings
- Display of Strings Using Different Formatting Techniques
- Array of Strings
- String Functions and Applications

BLOCK 2: Functions, Structures, Pointers and File Handling in C Unit 5: Functions

- Definition of a Function
- Declaration of a Function
- Function Prototypes
- The Return Statement
- Types of Variables and Storage Classes ○ Automatic Variables ○ External Variables ○ Static Variables ○ Register Variables
- Types of Function Invoking
- Call by Value
- Recursion

Unit 6: Structures and Unions

- Declaration of Structures
- Accessing the Members of a Structure
- Initializing Structures

- Structures as Function Arguments
- Structures and Arrays
- Unions
- Initializing an Union
- Accessing the Members of an Union
- Bit fields

Unit 7: Pointers

- Pointers and their Characteristics
- Address and Indirection Operators
- Pointer Type Declaration and Assignment
 - Pointer to a Pointer
 - Null Pointer Assignment
- Pointer Arithmetic
- Passing Pointers to Functions
 - A Function Returning More than

◦ String Input / Output Functions
 Input / Output • Standard Library Modules
 Functions

◦ Block Input / Output Functions **Block-4 Advanced Features in Python**

- Sequential Vs Random Access Files
- Positioning the File Pointer
- The Unbuffered I/O - The UNIX like File Routines

PYTHON PROGRAMMING (BLOCK-3&4)

Block-3 Introduction to Python Programming

One Value ◦ Function
 Returning a Pointer

- Arrays and Pointers
- Array of Pointers
- Pointers and Strings

Unit 8: File Handling

- File Handling in C Using File Pointers
 - Open a file using the function *fopen ()*
 - Close a file using the function *fclose()*
- Input and Output using file pointers
 - Character Input and Output in Files

Unit-9 Introduction to Python

- History of Python
- Need of Python

- Packages for Cross platform application of Python
- Getting started with Python
- Program structure in python
- Running the First program

Unit-10 Data Structures and control statements in Python

- Data Types
- Data Structures
 - Arrays
 - Linked Lists
 - Stacks
 - Queues
- Sequence, Selection and Iteration constructs
- Recursion

Unit-11 Functions and File Handling in

• Package Creation and Importing

◦ Formatted Input / Output

Block-4 Advanced Features in Python

Unit-13 Classes in Python

- Introduction to Object Oriented Paradigms
- Classes and instances
- Classes method calls
- Inheritance and Compositions
- Static and Class Methods
- Operator Overloading
- Polymorphism

Python

- Function definition and call
- Function Scope
- Arguments
- Function Objects
- Lambda Functions
- Anonymous Functions
- File Operations
- Creating. Opening and using files

Unit-12 Modules and Packages

- Module Creations and Usage
- Module Search Path
- Module Vs. Script

Unit-14 Exception Handling in Python Programming

- Default Exception Handler

- Catching Exceptions
- Iterators

MCS-208 Data Structures and Algorithms 4 Credits

Objectives

The learner should become well versed with Algorithms and various data structures. S/he should be able to use them appropriately as per need during development of programs. Also, the learner should know different sorting and searching techniques so that correct techniques can be used in different programs so that the complexity of the program does not increase due the sorting/search technique employed. The course also includes Advanced Data Structures which will enable learner to apply them in solving complex problems.

Block and Unit Structure

BLOCK 1: Introduction to Algorithms and Unit 4: Stacks

Data Structures

Unit 1: Analysis of Algorithms

- Mathematical Background
- Process of Analysis
- Calculation of Storage Complexity
- Calculation of Run Time Complexity

Unit 2: Arrays, Pointers and Structures

- Arrays and Pointers

Unit 5: Queues

- Sparse Matrices
 - Structures
 - Polynomials
 - Representation of Arrays
 - Row Major Representation
 - Column Major Representation
 - Applications
- ##### Unit 3: Lists
- Abstract Data Type-List
 - Array Implementation of Lists
 - Linked Lists-Implementation
 - Doubly Linked Lists-Implementation
 - Circularly Linked Lists-Implementation
 - Skip lists
 - Applications
- ##### Unit 6: Trees
- Abstract Data Type-Tree

Implementation of Tree

- Raise an exception
- User defined exception

Unit-15 Advanced Concepts

- Decorators
- Generators

- Abstract Data Type-Stack
- Implementation of Stack
 - Implementation of Stack using Arrays
 - Implementation of Stack using Linked Lists
- Algorithmic Implementation of Multiple Stacks
- Applications

- Abstract Data Type-Queue
- Implementation of Queue
 - Array Implementation
 - Linked List Implementation
- Implementation of Multiple Queues
- Implementation of Circular Queues
 - Array Implementation
 - Linked List Implementation
- Implementation of a circular queue
- Priority Queues
- Implementation of DEQUEUE
 - Array Implementation of a dequeue
 - Linked List Implementation of a dequeue

BLOCK 2: Stacks, Queues and Trees

- Co-routines

Unit-16 Data Access using Python

- Database concepts
- Creating database
- Querying Database

- Using SQL to get more out of database

- Tree Traversals
- Binary Trees
- Implementation of Binary Tree
- Binary Tree Traversals ○ Recursive Implementation of Binary Tree Traversals ○ Non Recursive Implementations of Binary Tree Traversals
- Applications

BLOCK 3: Graph Algorithms and Searching Techniques

Unit 7: Advanced Trees

- Binary Search Trees ○ Traversing a Binary Search Trees ○ Insertion of a node into a Binary Search Tree ○ Deletion of a node from a Binary Search Tree
- AVL Trees ○ Insertion of a node into an AVL Tree ○ Deletion of a node from and AVL Tree ○ AVL tree rotations ○ Applications of AVL Trees
- B-Trees ○ Operations on B-Trees ○ Applications of B-Trees
- Splay Trees ○ Splaying steps ○ Splaying Algorithm
- Red-Black trees ○ Properties of a Red-Black tree
- AA-Trees

Unit 8: Graphs

- Definitions
- Shortest Path Algorithms ○ Dijkstra's Algorithm ○ Graphs with Negative Edge costs ○ Acyclic Graphs ○ All Pairs Shortest Paths Algorithm
- Minimum cost Spanning Trees ○ Kruskal's Algorithm ○ Prims's Algorithm ○ Applications
- Breadth First Search
- Depth First Search
- Finding Strongly Connected Components

Unit 9: Searching and Sorting Techniques

- Linear Search
- Binary Search
- Applications
- Internal Sorting ○ Insertion Sort ○ Bubble Sort ○ Quick Sort ○ 2-way Merge Sort ○ Heap Sort
- Sorting on Several Keys
- External Sorting Algorithms

BLOCK 4: File Structures and Advanced Data Structures

- Unit 10: Hashing** ○ Introduction
 ○ Index Mapping ○ Collision Handling ○ Double Hashing ○ Load Factor and Rehashing

Unit 11: Advanced Data Structures

- Scapegoat Trees
- Tries
- Binary Tries
- X-Fast Tries
- Y-Fast Tries

Unit 12: File Structures

- Terminology
- File Organisation
- Sequential Files ○ Structure ○ Operations ○ Disadvantages ○ Areas of use
- Direct File Organisation
- Indexed Sequential File Organisation

SEMESTER - II

MCS-218 Data Communication and Computer Networks Credits: 4

The course introduces the fundamental concepts of data communication and Computer Networks. In the networking field, significant changes have taken place: (i) evolution of the Internet and wireless networks (ii) growth of networking services and applications. Network security has become very important topics because things are becoming digital and networked with each other. One extra unit has been added to cover the security topics. The objective of the revised courses is to reflect these changes besides explaining the basic principles of computer networking. A several new topics have been introduced in the revised course: Personnel Area network: Bluetooth and Zigbee, Cellular Networks : Architecture, Handoff, 3G,4G and 5G networks, Mobile IP, IPV6, Mobile Adhoc Networks, Wireless Sensor Networks, Internet of Things (IOT), Network Layer Security: IPSec, VPN, Securing TCP Connections: SSL, WLAN Security, Cyber Threats and Attacks and Counter Measures, Taxonomy of various Cyber Attacks, Virus, Worm and Trojan , DoS attack, DDOS attack, Phishing attacks, Malware, Ransom, vulnerabilities, Buffer Overflow, SQL Injection, Browser Vulnerabilities, OS vulnerabilities, Basics Computer Forensics, Recent Cyber Attacks and Firewalls and Intrusion Detection Systems.

Course Structure*

Block- 1 Introduction to Data Communication

Unit 1: Introduction to Internet

- Introduction
- Objectives
- What is the Internet?
ISP and Internet Backbone
Interconnection of ISPs
 - Taxonomy of Network
 - Standard Internet Protocols
 - Public Network & Private Network (Intranet)
 - Accessing the Internet
 - Telephone Network
 - Cable Network
 - Wireless Network
- Internet Services
Network Topology
Network Models
- OSI Model
- TCP/IP Model
- Summary
- Solutions/Answers

Unit 2: Data Transmission basics & transmission media

Introduction

Objectives

Data Communication Terminology

Channel

Baud

Bandwidth

Frequency

Modes of Data Transmission

Serial and Parallel Communication

Synchronous, Asynchronous and

Isochronous Communication

Simplex, Half Duplex and Full

Duplex Communication

Analog and Digital Data Transmission

Transmission Impairments

Attenuation

Delay Distortion

Noise

Signal to Noise ratio

Concept of Delays

Transmission Media and its

Characteristics

Guided media

Unguided media

Wireless Transmission

Microwave Transmission

Radio Transmission

Infrared and Millimeter Waves

Wireless LAN

Summary

Solutions/Answer

Unit 3: Data Encoding & multiplexing

Introduction
 Objectives
 Encoding
 Analog to Analog Modulation
 Analog to Digital Modulation
 Digital to Analog Modulation
 Digital to Digital Encoding
 Multiplexing
 Frequency Division Multiplexing
 Time Division Multiplexing
 Summary
 Solutions/Answers

Block- 2 Media Access Control and Data Link Layer

Unit 1: Data Link Layer Fundamentals

Introduction
 Objectives
 The services provided by the link layer
 Framing
 Error Correction and Detection

- Type of errors
- Single bit error
- Burst error
- Error Detection and Correction Techniques
- Parity Check
- Checksum Methods
- Cyclic Redundancy check

 DLC Protocols

- HDLC (High Level Data Link Control)
- PPP Protocol (Point to Point Protocol)

 Flow Control
 Summary
 Solutions/Answers

Unit 2: Retransmission Strategies

Introduction
 Objectives
 Stop & Wait ARQ
 Sliding window Protocols
 Piggybacking and Pipelining Concepts
 Go-Back-N ARQ (Automatic Repeat Request)
 Selective Repeat N.
 Summary
 Solutions/Answers
 Further Readings

Unit 3: Contention-based Media Access Protocols

Introduction
 Objectives
 Advantages of Multiple Access
 Sharing of Channel Resources Pure ALOHA
 Slotted ALOHA
 Carrier Sense Multiple Access(CSMA)
 CSMA with Collision Detection (CSMA/CD)
 Ethernet Frame Format (IEEE802.3)
 Summary
 Solutions/Answers
 Further Readings

Unit 4: Polling-based Media Access Control Protocols

- Introduction
- Objectives
- Characteristics of Wireless Link and Wireless Network
- Introduction to Wireless LAN
- Wireless LAN Architecture (IEEE802.11)
- Hidden Station and Exposed Station Problems • Wireless LAN Protocols:
 - MACA and MACAW
- IEEE 802.11 Protocol Stack
- The 802.11 Physical Layer
- The 802.11 MAC Sub-layer Protocol • Switching at Data Link Layer
- Personnel Area network: Bluetooth and Zigbee
- Cellular Networks: Architecture, Handoff
- 3G, 4G and 5G networks
- Summary
- Solutions/Answers
- Further Readings

Block- 3 Network Layer

Unit 1: Introduction to Layer Functionality and Design Issues

- Introduction ○ Objectives ○ Connection Oriented vs. Connection-less Services

- Connection-oriented Services
- Connection-less Services
- Implementation of the Network Layer Services
 - Packet Switching
 - Implementation of Connection-oriented Services
 - Implementation of Connection-less Services
- Comparison between Virtual Circuit and Datagram Subnet
- Addressing
 - Hierarchical Versus Flat Address
 - Static vs. Dynamic Address
 - IP Address
- Concept of Congestion
- Routing
 - Main Issues in Routing
 - Classification of Routing Algorithm
- Summary
- Solutions/Answers
- Further Readings

Unit 2: Routing Algorithms

Introduction

Objectives

Flooding

Shortest Path Routing Algorithm

Distance Vector Routing

Comparison

- The Count-to-Infinity Problem

- Link State Routing
- Hierarchical Routing
- The Internet Protocol (IP)
 - IPv4 addressing
 - Datagram Format
 - IPv6
 - IP Datagram Fragmentation
 - Internet control message protocol
 - Dynamic host configuration protocol
 - IP Security
- Routing with Internet
 - Inter Autonomous System
 - Routing in the Internet: RIP & OSPF
 - Inter Autonomous System
 - Routing BGP

- Multicast Routing
- Mobile IP
- Summary
- Solution/Answers
- Further Readings

Unit 3: Congestion Control Algorithms

- Introduction
- Objectives
- Reasons for Congestion in the network
- Congestion Control vs. Flow Control
- Congestion Prevention Mechanism
- General Principles of Congestion Control
- Open Loop Control
 - Admission Control
 - Traffic Policing and its Implementation
 - Traffic Shaping and its Implementation
 - Leaky Bucket Shaper
 - Token Bucket Shaper
 - Difference between Leaky Bucket Traffic Shaper and token Bucket Traffic Shaper
- Congestion Control in Packetswitched Networks
- Summary
- Solution/Answers
- Further Readings

Unit 4: Emerging Networking Technology

- Mobile Adhoc Networks
- Wireless Sensor Networks
- Internet of Things(IOT)

Block- 4 Transport Layer and Application Layer Services

Unit 1: Transport Services and Mechanism

- Introduction
- Objectives
- Transport Services
 - Types of Services
 - Quality of Services
 - Data Transfer
 - Connection Management
 - Expedited Delivery

- Elements of Transport Layer Protocols
 - Addressing
 - Multiplexing
 - Flow Control and Buffering
 - Connection Establishment
 - Crash Recovery
- Summary
- Solutions/Answers
- Further Readings

Unit 2: TCP/UDP

- Introduction
- Objectives
- Services Provided by Internet Transport Protocols
 - TCP Services
 - UDP Services
- Introduction to UDP
- Introduction to TCP
- TCP Segment Header
- TCP Connection Establishment
- TCP Connection Termination
- TCP Flow Control
- TCP Congestion Control
- Remote Procedure Call
- TCP in wireless environments
- Summary
- Solutions/Answers
- Further Readings

Unit 3: Network Security I

- Introduction
- Objectives • What is Internet Security?
- Principles of Cryptography
- Symmetric Key Cryptography
- Public Key Cryptography

- RSA Public Key Algorithm • Application of Public Key Cryptography (Digital Signature)
- Management of Public Keys
- Kerberos
- Network Layer Security: IPSec, VPN
- Securing TCP Connections: SSL
- WLAN Security
- Summary
- Solutions/Answers
- Further readings

Unit 4: Network Security-II

- Introduction
- Objectives
- Cyber Threats and Attacks and Counter Measures
- Taxonomy of various Cyber Attacks
- Virus, Worm and Trojan, DoS attack, DDOS attack, Phishing attacks, Malware, Ransom
- vulnerabilities
- Buffer Overflow
- SQL Injection
- Browser Vulnerabilities
- OS vulnerabilities
- Basics Computer Forensics
- Recent Cyber Attacks
- Firewalls and Intrusion Detection Systems
- Summary
- Solutions/Answers
- Further Readings

MCS-219

Object Oriented Analysis and Design

4 Credits

Object oriented analysis and design is a popular paradigm of analysis and design of the systems. This Course is designed to help in learning object oriented analysis and design concepts. This Course is having coverage of UML diagrams and will help in developing understanding in the area of system analysis and design concepts using object-oriented approach. This Course will cover different aspects of OOAD with explaining object modeling dynamic modeling and functional modeling. The topics covered in the course include:

Object Oriented Modeling and UML

Introduction to Object Oriented Modelling: OOT Object Oriented Modeling, Characteristics Object Oriented Modeling (Class and Objects, Links and Association, Generalization and Inheritance), An Object Model, Benefits of OO Modeling, Introduction to OOAD tools

Object Oriented Analysis: Object Oriented Analysis, Problem Statement: an Example, Differences between Structured Analysis and Object Oriented Analysis, Analysis Techniques (Object Modeling, Dynamic Modeling, Functional Modeling), Adding Operations, Analysis Iteration

Using UML: UML: Introduction, Object Model Notations: Basic Concepts, Structural Diagrams (Class, Object, Composite, Package, Component, Deployment), Behavioural Diagrams (Use Case, Communication, Sequence, Interaction Overview, Activity, State), Modelling with Objects

Object Oriented Design

System Design: System Design: An Object-Oriented Approach, Breaking into Subsystems, Concurrency Identification, Management of data store, Controlling events between Objects, Handling Boundary Conditions

Object Design: Object Design for Processing, Object Design Steps, designing a Solution, Choosing Algorithms, Choosing Data Structures, Defining Classes and delegation of Responsibilities to Methods

Advance Object Design: Control and its Implementation (Control as a State within Program, Control as State Machine Engine, Control as Concurrent Task), Inheritance Adjustment,

Association: Design, Object Representation, Design Optimization, Design Documentation

Modeling

Object Modeling: Advance Modeling Concepts (Aggregation, Abstract Class), Multiple Inheritance, Generalization as an Extension, Generalization as a Restriction, Metadata, Constraints, An Object Model

Dynamic Modeling: Events, State and State Diagram, Elements of State Diagrams, Examples of State Diagrams, Advance Concepts in Dynamic Modeling, Concurrency, A Dynamic model Functional Modeling: Functional Models, Data Flow Diagrams, Features of a DFD, Design flaws in DFD, A Functional model, Relationship between Object, Dynamic, and Functional Models **Implementation**

Implementation Strategies: Implementation (Using Programming Languages, Using Database System), Unidirectional Implementation, Bi-directional Implementation, implementing associations, Implementing Constraints, Implementing State charts, Persistency

Object Mapping with Databases: Relational Database Schema for Object Modes, Object Classes to Database Tables, Mapping Associations to Tables, Mapping Generalizations to Tables, Interfacing to Database, Object Mapping with Databases: an Example.

Course Structure*

Block 1: Object Oriented Analysis and UML

Unit 1: Introduction to Object Oriented Modeling

- Introduction to Object Orientation
- Basic Philosophy of Object Orientation
- Principals of Object Orientation
- Abstraction
- Encapsulation
- Inheritance

- Polymorphism
- Basic Constructs in Object Orientation
 - Class and Objects
 - Links and Association
 - Generalization and Special
- Identifying Class and Object
- Benefits of Object Orientation
- Introduction to OOA & Design Tools

Unit 2: Structural Modeling using UML

- Introduction to UML
- Basic Structural Modeling
 - Classes ○ Relationships,
 - Common Mechanisms ○ Class Diagram
- Advanced Structural Modeling
 - Advance Classes ■ Advanced Relation
 - Interference Type and Roles
- Packages,
- Instance and Object Diagrams

Unit 3: Behavioral Modeling using UML

- Basic Behavioral Modeling
- Interactions,
- Use Cases and Use Case Diagram
- Interaction Diagram
- Activity Diagram

Unit 4: Advanced Behavioral Modeling using UML

- Events and Signals
- State Machines
- Process and Threads
- Time and Space
- State Chart Diagram

Unit 5: Architectural Modeling

- Components
- Deployment
- Collaboration
- Component Diagrams
- Deployment Diagrams

Block 2: Modeling

Unit 1: Object Modeling

- Advanced Modeling Concepts
 - Aggregation
 - Abstract Class
- Multiple Inheritance
 - Generalization and Specialisation

- Meta Data and Keys
- Integrity Constraints
- An Object Model

Unit 2: Dynamic Modeling

- Events
- State and State Diagram
- Elements of a State Diagram
- Advanced Concepts in Dynamic Modeling
- Concurrency
- A Dynamic Model

Unit 3: Functional

Modeling •

Functional Models

- Data Flow Diagrams
- Features of a DFD
 - Processes ○ Data Flows ○ Actors ○ Data Stores ○ Constraints ○ Control Flows
- Design Flaws in DFD
- A Sample Functional Model • Relation of Functional to Object and Dynamic Model

Block 3: Object Oriented Design

Unit 1: Basics of System Design

- OOA to OOD
- System Design: An Object-Oriented Approach
- Breaking into Subsystems
- Concurrency Identification
- Management of a Data Store
- Controlling Events Between Objects
- Handling Boundary Conditions

Unit 2: Object Design

- Object Design for Processing
- Object Design Steps
- Choosing Algorithms ○
 - Selecting Data Structure ○
 - Defining Internal Classes and Operations ○
 - Assigning Responsibility for Operation
- Implementation of Control ○ State as Location within a Program ○ State Machine Engine ○ Control as Concurrent Tasks
- Adjustment of Inheritance ○ Rearranging Classes and Operations ○ Abstracting Out Common Behavior
- Design of Associations ○ Analyzing Association Traversal ○ One-way Associations ○ Two-way Associations

Unit 3: Advance Object Design

- Control and its Implementation ○ Control as a State within Program ○ Control as a State Machine Engine ○ Control as Concurrent Task
- Inheritance Adjustment
- Association: Design
- Object Representation
- Design Optimization
- Design Documentation

Block 4: Implementation

Unit 1: Implementations Strategies -1

- Mapping Design to Code
- Creating Class Definition from Class

Diagram

- Implementing Associations
- Unidirectional Implementations ○ Optional Associations ○ One-to-One Associations ○ Associations with Multiplicity

‘Many’

- Bi-directional Implementations ○ One-to-One and Optional Associations ○ One-to-Many Associations ○ Immutable Associations

Unit 2: Implementation Strategies-2

- Creating Methods from Collaboration Diagram
- Implementing Constraints
- Implementing State Charts
- Persistency

Unit 3: Objects Mapping With Databases

- Relational Database Schema for Object Modes ○ General DBMS Concepts ○ Relational DBMS Concepts ○ RDBMS Logical Data Structure
- Object Classes to Database Tables ○ Extended Three Schema

Architecture for Object Models ○ The use of Object IDs ○ Mapping Object Classes to Tables

- Mapping Associations to Tables ○ Mapping Binary Associations to Tables
- Mapping Many-to-Many Association to Tables ○ Mapping Ternary Associations to Tables

Main objective of the Course is to introduce concepts, tools/technologies and programming to develop distributed secure, reliable and scalable Web Application using J2EE Technologies application. This course discusses some commonly used design patterns, servlet, JSP, Spring Boot & Hibernate (ORM), and Web Security. The topics covered in the course include:

Introduction to Advance Java (J2EE) J2EE Architecture and Design patterns (MVC, Repository Design pattern, Singleton, Factory, etc.), Building java Application JAR and WAR and deployment in tomcat.

Introduction to Servlets, Http Protocol & Http Methods, Web Server & Web Container, Servlet Architecture, Servlet Life Cycle, Steps to create a Servlet, Servlet Communication (Servlet- Browser, Web-component,) , Session Management, Database Connectivity in Servlet, Java Server Pages(JSP) Overview, JSP Life Cycle, JSP API, Components of JSP(Directives, Scripting, Action), JSP Implicit Objects, JSP Standard Tag Library (JSTL), Exception handling using JSP, Database Connectivity in JSP.

Introduction to J2EE Frameworks, discuss about various Frameworks available for J2ee Development (Struts, Hibernate, Spring)- Maven and Introduction of Annotation. Spring MVC- Configuration, Create, Read, Update, and Delete (CRUD) Application. Spring MVC with Bootstrap CSS- Configuration of Bootstrap in Application and Apply custom CSS in pages.

Spring Boot & Hibernate (ORM) - Introduction to Spring boot, Configuration of Hibernate (ORM)

CRUD Application using spring boot and Hibernate.

Web Security- Spring Security configuration, Custom login using Security, Role based login.

- Mapping Generalizations to Tables
- Interfacing to Databases

Course Structure*

Block 1: Web Application

Development using J2EE

Unit 1: Introduction to J2EE, Architecture and Design pattern

- Web Server & Web Container,
- Introduction to J2ee
- Design Patters
 1. MVC
 2. Repository Design pattern
 3. Singleton
 4. Factory
- Building java Application JAR and WAR and deployment in tomcat

Unit 2: Basics of Servlet

- Introduction to Servlets
- Http Protocol & Http Methods
- Servlet Architecture

- Servlet Life Cycle
- Creating a Servlet
- Servlet Communication (Servlet-Browser and Webcomponent)

Unit 3: Session Management and Database Connectivity in Servlet

- Session Management
- Database Connectivity in Servlet,
- Servlet Communication (Servlet-Browser, Webcomponent)
- Servlet Collaboration
- Session Management
- Database Connectivity

Unit 4: JSP

- JSP Overview
- JSP Life Cycle
- JSP API

- Components of JSP (Directives, Scripting, Action)
- JSP Implicit Objects
- An Introduction to JSP Standard Tag Library (JSTL)
- Exception handling using JSP
- Database Connectivity

Block 2: Frameworks for J2EE

Unit 5: Introduction to J2EE

Frameworks

- Introduction of Struts
 - Introduction of Spring including Boot and MVC
 - Introduction of Hibernate with Java Persistence API (JPA)
 - Introduction of Annotation
- Unit 6: Discuss about various Frameworks available for J2EE Development (Struts, Hibernate, Spring)**
- Struts: Features
 - Spring Boot and MVC: features
 - Hibernate with JPA: Features
 - Compare amount these frameworks
 - Maven: Introduction., Overview and configuration
 - Create First Project using Maven

Unit 7: Spring MVC

- Setting up Development Environment for Spring MVC
- First Hello World Project using Spring MVC
- Inversion of Control (IoC) and Dependency Injection
- Creating Controllers and Views
- Request Params and Request mapping
- Form Tags and Databinding
- Form Validation

Unit 8: Spring MVC with Bootstrap CSS

- Configuration of Bootstrap in Spring Application
- Apply custom CSS in pages
- Setting UP Database using Hibernate
- Create, Read, Update, and Delete (CRUD)

- CRUD examples in Spring MVC and Hibernate

Block 3: Spring Boot and Hibernate (ORM)

Unit 9: Introduction to Spring boot

- Spring Boot: Overview
- Spring Boot Dev Tools and Spring Boot Actuator
- Spring boot- Application Properties
- Running Spring Boot Apps from command line

Unit 10: Configuration of Hibernate (ORM)

- Hibernate Overview
- Hibernate Configuration with Annotation
- REST (REST stands for

Representational State Transfer)
JPA Overview

- Creating JPA DAO implementation for REST API
- Hibernate CRUD (Create, Read, Update, and Delete)Features

Unit 11: CRUD Application using

Spring boot and Hibernate

- Create records using Spring Boot and Hibernate
- Read records using Spring Boot and Hibernate
- Update records using Spring Boot and Hibernate
- Delete records using Spring Boot and Hibernate

Block 4: Web Security

Unit 12: Spring Security configuration

- Introduction to Web Securities ○ Introduction of Java Cryptography Architecture (JCA) ○ Introduction of Java Secure Socket Extension (JSSE)
- Issues and Challenges of Web Security
- Spring Security Overview
- Java based configuration
- Create Spring Initializer class
- Create Controller and View
- Run Application

Unit 13: Custom login using Security

- Custom login form creation
- Spring Config for Custom Login Form
- Create Request mapping and building Custom Login Form

- Testing Custom Login Form
- Adding Logout Support

Unit 14: Role based login

- Display User Id and Roles Overview
- Roles based login Example
- Restrict Access based on Roles
- Testing the Application
- Cross Site Request Forgery (CSRF)

MCS-221 Data Warehousing and Data Mining 4 Credits

The course objectives are:

- To understand the underlying concepts of Data Warehousing
- To identify the components of the Data Warehouse Architecture
- To know the difference between the Data Warehouse and Data Marts
- To understand the Data Warehouse Development Life Cycle
- To elucidate the dimensional modeling techniques
- To understand the ETL, OLAP concepts and other evolving trends
- To learn data mining concepts and understand association rules mining
- To discuss classification algorithms, learn how data is grouped using clustering techniques
- To develop the abilities of critical analysis to data mining systems and applications
- To implement practical and theoretical understanding of the technologies for data mining
- To understand the strengths and limitations of various data mining models The topics covered in the course include:

Introduction to Data Warehousing, Evolution of Data Warehousing, Features of Data Warehousing, Benefits of Data Warehousing, Data Granularity, Metadata, Data Warehousing Architecture, Data Warehouse and Data Marts, Building Data Marts, Issues in building data marts, Data Warehouse Schema, Dimensional Modeling, The Star Schema, The Snowflake Schema, Aggregate Tables, Fact Constellation Schema, Dimensional Modeling, Extraction, Transformation and Loading (ETL) process, OLAP and Data Warehousing, OLTP and Data Warehousing, Trends in Data Warehousing

Introduction to Data Mining Systems, How Data Mining Works, Classification of Data Mining Systems Issues, Applications of Data Mining, Data Mining Tools, Issues in Data Mining, Data Preprocessing – Cleaning, Integration, Reduction, Transformation and Discretization, Data similarity and dissimilarity measures, Mining Frequent Patterns, Associations, Classification using Frequent Patterns, Decision Tree Induction, Bayesian Classification, Rule Based Classification, Classification by Back Propagation, Support Vector Machines, K- nearest Neighbor classification, Clustering, Major Clustering Methods, Partitioning Methods, Hierarchical Methods, Density Based Methods, Grid Based Methods, Hierarchical Clustering,

Outlier Detection, Text and Web Mining.

Course Structure*

BLOCK 1: DATA WAREHOUSE FUNDAMENTALS AND ARCHITECTURE

UNIT 1: Fundamentals of Data Warehouse

- Introduction to Data Warehousing
- Evolution of Data Warehousing
- Data Warehousing Concepts
- Online Transaction Processing Systems
- Differences between OLTP Systems and Data Warehouse
- Characteristics of Data Warehouse
- Data Granularity
- Metadata and Data Warehousing
- Functionality of Data Warehouse
- Advantages of Data Warehouse
- Applications of Data Warehouse
- Concerns in Data Warehouse
- Types of Data Warehouses ○ Enterprise Data Warehouse ○ Operational Data Store ○ Data Mart

Unit 2: Data Warehouse Architecture

- Introduction to Data Warehouse Architecture
- Characteristics of Data Warehouse Architecture
- DW Architecture Goals
- Components of Data Warehouse ○ Load Manager ○ Warehouse

Manager ○ Query Manager • Data Mart

- Building Data Marts
- DW and Data Marts
- Issues in Building Data Marts
- Co-existence of DW and Data Mart
- Planning and Requirements ○ Planning Data Warehouse and Key Issues ○ Planning and Project Management in constructing Data Warehouse ○ Data Warehouse Development Life Cycle ○ Methodologies - Top- Down, Bottom-Up and Hybrid Development Methodology

Unit 3: Dimensional Modeling

- Introduction to Dimensional Modeling and its Strengths
- Identifying Facts and Dimensions
- Star Schema
- Pros and Cons of Star Schema
- Snowflake Schema
- Pros and Cons of Snowflake Schema
- Aggregate Tables ○ Need for Building Aggregate Fact Tables ○ Limitations of Aggregate Fact Tables
- Fact Constellation Schema ○ Aggregate Fact Tables and Derived Dimension Tables
- Pros and Cons of Fact Constellation Schema

BLOCK 2: ETL, OLAP AND TRENDS

Unit 4: Extract, Transform and Loading

- Overview of ETL
- ETL requirements and steps
- Data Extraction
- Extraction Methods - Logical
Extraction Methods and Physical
Extraction Methods
- Data Transformation
- Basic Tasks in Transformation
- Major Data Transformation Types
- Data loading; Data Loading
Techniques
- Data Quality

Unit 5: Introduction to Online Analytical Processing

- Need for OLAP
- Characteristics of OLAP
- OLAP and Multidimensional Analysis
 - Multidimensional Logical Data
Model and its Users ◦
Multidimensional Structure ◦
Multidimensional Operations
- OLAP Functions • Data Warehouse
and OLAP:
Hypercube & Multi-cubes
- OLAP Applications
- Steps in the OLAP Creation Process
- Advantages of OLAP
- OLAP Architectures - MOLAP,
ROLAP, HOLAP, DOLAP

Unit 6: Trends in Data Warehouse

- Data Lakes
Complex Data Marts
- Cloud Data Warehousing
- Real Time Data Warehousing
- Data Warehousing and Hadoop
- Data Warehouse Automation

BLOCK 3: DATA MINING FUNDAMENTALS AND FREQUENT PATTERN MINING

Unit 7: Data Mining – An Introduction

- Introduction
- Data Mining – From What Kind of
Data ◦ Relational Databases ◦ Data
Warehouses ◦ Transactional Databases

◦ Advanced Data and Informational
Systems

- How does Data Mining Works?
- Classification of Data Mining Systems
- Applications of Data Mining
- Data Mining and Data Warehousing
- Data Mining Tools
- Major Issues in Data Mining

Unit 8: Data Preprocessing

- Introduction
- Data Preprocessing Overview
- Data Cleaning ◦ Missing Values ◦
Noisy Data ◦ Data Cleaning as a
Process
- Data Integration and Transformation ◦
Data Integration ◦ Data
Transformation
- Data Reduction ◦ Data Cube
Aggregation ◦ Attribute Subset
Selection ◦ Dimensionality
Reduction ◦ Numerosity
Reduction
- Discretization and Binarization •
Measures of Similarity and
Dissimilarity-Basics

Unit 9: Mining Frequent Patterns and Associations

- Problem Definition
- Frequent Item Set Generation
- The APRIORI Principle
- Support and Confidence Measures
- Association Rule Generation
- APRIORI Algorithm: Finding
Frequent Itemset Using Candidate
Generation
- Generating Association Rules from
Frequent Item set
- Improving the efficiency of Apriori
- Correlation Analysis
- From Association Analysis to
Correlation Analysis

BLOCK 4: CLASSIFICATION, CLUSTERING AND WEB MINING

Unit 10: Classification

- Introduction
- Classification: Problem Definition

- General Approaches to solving a classification problem
- Evaluation of Classifiers
- Classification techniques
- Decision Trees-Decision tree Construction
- Methods for Expressing attribute test conditions
- Measures for Selecting the Best Split
- Algorithm for Decision tree Induction
- Bayesian Classification ○ Bayes' Theorem ○ Naive-Bayesian Classification ○ Bayesian Belief Networks
- Support Vector Machines ○ The Case when the data are linearly separable
 - The Case when the data are linearly inseparable

Unit 11: Clustering

- Clustering: Problem Definition
- Clustering Overview
- Categorization of Major Clustering Methods ○ Partitioning Method
 - Hierarchical Method

- Density-based Method ○ Grid-Based Method ○ Model-Based Method ○ Constraint-based Method

- Partitioning Method ○ K-Means Algorithm
 - K-Medoids
- Hierarchical Clustering ○ Agglomerative Method ○ Divisive Method
- Key Issues in Hierarchical Clustering
- Strengths and Weakness
- Outlier Analysis – Outlier Detection methods

Unit 12: TEXT AND WEB MINING

- Text and Web Mining: Introduction
- Text Data Analysis and Information Retrieval
- Dimensionality Reduction for Text
- Text Mining Approaches
- Web mining
- Web content mining
- Web structure mining
- Mining Multimedia Data on the Web
- Automatic Classification of Web Documents
- Web usage mining

MCSL-222 OOAD and Web Technologies Lab 2 Credits

Main objective of this laboratory course is to provide hands on exercises to the learners based on Object Oriented Analysis and Design & Web Technologies Courses.

Lab Sessions:

- There will be 20 practical sessions (3 hours each) of which 10 sessions will be on OOAD and 10 sessions will be on Web Technologies.
- The practice problems for all 20 sessions will be listed session-wise in the lab manual.

MCSL-223 Computer Networks and Data Mining Lab 2 Credits

Main objective of this laboratory course is to provide hands on exercises to the learners based on Computer Networks and Data Mining Courses.

Lab Sessions:

- There will be 20 practical sessions (3 hours each) of which 10 sessions will be on Computer Networks and 10 sessions will be on Data Mining.
- The practice problems for all 20 sessions will be listed session-wise in the lab manual.

SEMESTER - III

MCS-224 Artificial Intelligence and Machine Learning (CREDITS-4)

The course relates to the conceptual understanding of the Artificial Intelligence and Machine Learning. Generally, Artificial Intelligence (AI) is considered as the discipline, to deal with the solution of the hard and insolvable problems using reasonable amount of time, by exploiting the knowledge of the problem domain. In view of the significance of knowledge in AI, in this course, a number of knowledge representation formalisms are introduced. The formalisms discussed include Propositional Logic, First Order Predicate Logic, Rule-based systems, Semantic Networks and Frames. Further, the course introduces the various concepts of Machine learning, Viz. Supervised learning, Unsupervised Learning and their respective application areas. Currently these two fields i.e. Artificial Intelligence and Machine Learning are in high demand, the course will help the learners to build the understanding of these fields.

Course Structure*

Block-1 Artificial Intelligence - Introduction:

Unit-1 Introduction to Artificial Intelligence -

What is AI? Examples of AI systems, Approaches to AI, Brief history of AI, Comparison Between Artificial intelligence, Machine Learning, and Deep Learning, Intelligent Agent: stimulusresponse agents. components of intelligence.

Unit-2 Problem Solving using Search – Single agent search: Introduction to State Space Search, Statement of Search problems: state space graphs, Searching explicit state spaces. Feature based state spaces. Problem types, examples (puzzle problem, n-queen, the road map) Two agent search: Adversarial search: Two agent games (alpha-beta pruning). MinMax Search.

Unit-3 Uninformed and Informed Search

–
Uninformed Search: Formulating the state space, iterative deepening, bidirectional search. Informed Search Strategies: Using evaluation functions. A* & AO* ,

admissibility of A* , Iterative deepening A* , recursive best first search.

Unit-4 Predicate and Propositional Logic

–
Propositional logic, syntax, semantics, semantic rules, terminology - validity, satisfiability. interpretation, entailment, proof systems. Propositional Logic inference rules, natural deduction, propositional resolution.

Block-2 Artificial Intelligence - Knowledge Representation:

Unit-5 First Order Logic - First Order Logic: Motivation, Syntax, Interpretations, semantics of quantifiers, Entailment in FOL, Interpretation, Inference in FOL: First Order resolution. Conversion to clausal form. Unification. Most general unifier. Resolution with variables Proving validity.

Unit-6 Rule based Systems and other formalism -

Rule Based Systems: Forward chaining.
Backward chaining. Conflict resolution.
Semantic nets, Frames, Scripts.

Unit-7 Probabilistic Reasoning

Reasoning with uncertain information
Review of Probability Theory,
Introduction to Bayesian Theory,
Bayes' Networks, Probabilistic

Inference, Basic idea of inferencing with Bayes networks. Other paradigms of uncertain reasoning.
Dempster-Scheffer Theory

Unit-8 Fuzzy and Rough Set Fuzzy Reasoning Introduction to Fuzzy sets, Fuzzy set representation, Fuzzy inferences, Rough Set Theory

Block-3 Machine Learning - I:

Unit-9 Introduction to Machine Learning Methods – Introduction to Machine Learning, Techniques of Machine Learning, Reinforcement Learning and algorithms, Deep Learning and its Algorithms, Ensemble Methods.

Unit-10 Classification – Understanding of Supervised Learning, Introduction to Classification, Classification Algorithms: Naïve Bayes, K-NN, Decision Trees, Logistic Regression, Support Vector Machines.

Unit-11 Regression – Introduction to Regression, Regression algorithm Linear Regression and Polynomial Regression, Support Vector Regression

Unit-12 Neural Networks and Deep Learning:

Overview of Artificial Neural Networks, Multilayer Feedforward Neural networks with Sigmoid activation functions; Back propagation Algorithm; Representational abilities of feed forward networks, Feed forward networks for Classification and Regression, Deep Learning

Block-4 Machine Learning - II:

Unit-13 Feature selection and Extraction: Introduction to Feature Selection and Extraction, Dimensionality Reduction, Principal Component Analysis, Linear Discriminant Analysis, Singular Value Decomposition.

Unit-14 Association Rules – Introduction to Pattern search and its algorithms: Apriori Algorithms. and its variants, FP Tree Growth, Pincer Search.

Unit-15 Clustering – Introduction to Clustering, Types of Clustering, Partition Based, Hierarchical Based, Density Based Clustering Techniques, Clustering algorithms: KMeans, Agglomerative and Divisive, DBSCAN, Introduction to Fuzzy Clustering.

Unit – 16 Machine Learning Programming using Python

Implementations of various algorithms learned in different units of this course

MCS-225 Accountancy and Financial Management (Credits:4)

Financial Management and Accountancy course is aimed at making students aware of the basic accounting procedures and financial management processes. The central purpose of accounting is to make possible the periodic matching of costs (efforts) and revenues (accomplishments). The course describes four major topics: Accounting System, Understanding and Analysis of Financial Statements, Financial Management and Decisions and Working Capital Management. Accounting System deals with the framework of accounting. The focus is on scope and function of accounting in modern business. Understanding and Analysis of Financial Statements deals with preparation of final accounting statements, which includes preparation and analysis of Profit and Loss A/c. In this topic we also discuss ratio analysis. Ratio analysis is one of the most widely used analytic tool for financial analysis. Financial Management and Decisions deals with various aspects of financial management. Working Capital Management topic deals with the various sub components of working capital, which includes cash and treasury management.

The main objective of cash management is to maintain an optimum level of cash balance.

The objectives of this course are:

- Understand how debit and credit are determined for business transactions.
- Understanding the basics of business entries.
- Understanding Trial Balance and the techniques to prepare it.
- Understanding the preparation process of final accounts.
- Understanding and analysing the information contents of final accounts.
- Understand what gives money its time value.
- Understand how the various factors influence working capital requirements.
- Understand the various methods of computing working capital.
- Preparation of cash budget.
- Understand the role and function of treasury management.
- Understand the need for establishing sound credit policy along with NPA management.
- Understand the process for managing inventory.

Course Structure*

Block 1: Accounting System

Unit 1: Accounting and its Functions

- Introduction
- Objectives
- The Scope of Accounting
- The Emerging Role of Accounting
- Accounting as an Information System
- The Role and Activities of an Accountant
- Accounting Personnel
- The Nature of the Accounting Function
- The Organisation for Accounting and Finance
- Summary
- Keywords
- Self-Assessment Questions/Exercises
- Further Readings

Unit 2: Accounting Concepts and Standards

- Introduction
- Objectives
- The Accounting Framework
- Accounting Concepts

- Accounting Standards
- The Changing Nature of Generally Accepted Accounting Principles
- Attempts towards Standardisation
- Accounting Standards in India
- International Financial Reporting Standard(IFRS)
- Summary
- Keywords
- Self-Assessment Questions/Exercises
- Further Readings

Unit 3: Basic Accounting Process: Preparation of Journal, Ledger, Trial Balance and Bank Reconciliation Statement

- Introduction
- Objectives
- Accounting Equation
- Classification of Accounts
- Definitions of Journal and Ledger
 - The Journalising Process
 - Ledger Posting
 - Balancing an Account
- Trial Balance
- Objectives of Preparing Trial Balance
 - The Total Method of Preparing the Trial Balance
 - The Balance Method of Preparing the Trial Balance

- The Limitations of Trial Balance
- The Accounting Cycle
- Bank Reconciliation Statement ○ Causes of differences in bank balance as per cash book and passbook
- Utility of bank reconciliation statement
- Preparation of bank reconciliation statement
- Keywords
- Summary
- Solutions /Answers
- Further Readings

Block 2: Understanding and Analysis of Financial Statements

Unit 1: Preparation and Analysis of Final Accounts

- Introduction
- Objectives
- Trading Account ○ Opening/Closing Stock ○ Net Purchases ○ Direct Expenses ○ Net Sales
- Profit and Loss Account
- Difference between Trading and Profit & Loss Account
- Balance Sheet
- Constructing a Balance Sheet
- Classification of Balance Sheet's Items
- Adjustment Entries ○ Closing Stock
- Depreciation ○ Bad Debts ○ Provision for Bad and Doubtful Debts ○ Outstanding Expenses (Assets) ○ Prepaid Expenses (Assets) ○ Accrued Income ○ Income Received in Advance (Liability)
- Forensic Accounting
- Summary
- Keywords
- Solutions/Answers
- Further Readings

Unit 2: Cash Flow Statement

- Introduction
- Objectives
- Statements of Changes in Financial Positions (SCFP)
- Analysing Changes in Working Capital
- Fund Flow Statement
- Sources of Funds
- Uses (Applications) of Funds
- Preparation of Fund Flow Statement
- Cash Flow Statement
- Sources and Uses of Cash • Preparation of Cash Flow Statement and analysis
- Summary
- Keywords
- Self-Assessment Questions
- Further Readings

Unit 3: Ratio Analysis

- Introduction
- Objectives
- Categories of Ratios ○ Long-term Solvency Ratios ○ Liquidity Ratios (Short-term Solvency Ratios) ○ Activity or Turnover Ratios ○ Profitability Ratios ○ Market Test Ratios
- Utility of Ratio Analysis
- Diagnostic Role of Ratios
- Application of Formulas
- Summary
- Self-Assessment Questions/Exercises
- Solutions/Answers

Unit 4: Reading and Interpretation of Financial Statements

- Introduction
- Objectives
- Annual Report
- Financial statements and information gap
- Analysis of Profit and Loss A/c
- Analysis of Cash Flow Statement
- Analysis of Balance Sheet
- Techniques of financial statement analysis
- Summary

- Self-Assessment Questions/Exercises
- Solutions/Answers

Block 3: Financial Management and Decisions

Unit 1: Introduction to Financial Management

- Introduction
- Objectives
- Evolution of Financial Management
- Significance of Financial Management
- Principles of Financial Management
 - Investment Decision
 - Financing Decision
 - Dividend Decision
 - Liquidity Decision
- Objectives of Financial Management
- Economic Profit vs. Accounting Profit
- Agency Relationship
 - Problems Related with Agency Relationship
 - Costs of the Agency Relationship
- The Changing Financial Landscape
- Organisation of Financial Management
- Tasks and Responsibilities of Modern Financial Manager
- Summary
- Self-Assessment Questions/Exercises
- Solutions/Answers

Unit 2: Time Value of Money

- Introduction
- Objectives
- Determining the Future Value
 - Shorter Compounding Period
 - Effective vs. Nominal Rates
 - Continuous Compounding
- Annuity
- Summary
- Self-Assessment Questions/Exercises
- Solutions/Answers

Unit 3: Cost of Capital

- Introduction
- Objectives
- Significance of the cost of capital
- Opportunity cost of capital
- Determining component cost of capital
 - Cost of debt
 - Cost of preference capital
 - Cost of equity capital
- Weighted average cost of capital
- Summary
- Self-Assessment Questions

Unit 4: Investment Decision Methods

- Introduction
- Objectives
- The Investment Problem
- Capital Investment and Firm's Value
 - Stages in Capital Budgeting Process
 - Importance of Capital Investment Decisions
 - Types of Investment Decisions
- Investment Evaluation Criteria
 - Non-Discounts Cash Flow techniques
 - Discounted Cash Flow techniques
- Summary
- Self-Assessment Questions/Exercises
- Solutions/Answers

Unit 5: Working Capital Decisions

- Introduction
- Objectives
- Characteristics of Current Assets
- Operating Cycle Concepts
- Factors Influencing Working Capital Requirements
- Sources of Working Capital
- Strategies of Working Capital Management
- Estimating Working Capital Requirement
- Summary
- Self-Assessment

Questions/Exercises

• Objectives of Inventory

• **S**
MCS-226 Data Science and Big Data Credit :4

This course introduces the students to the concepts of data science and big data, its architecture and a programming technique R that can be used to analyse big data.

Block 1: Basics of Data Science

Unit 1: Introduction to Data Science Definition
of Data Science Data
Answers

Block 4: Working Capital Management

Unit 1: Cash and Treasury Management

- Introduction
- Objectives
- Facets of Cash Management ○
Motives for Holding Cash ○ Cash
Planning ○ Determining Optimum
Cash Balance
- Methods of Cash Flow Budgeting
- Investing Surplus Cash
- Cash Collection and Disbursements
- Treasury Management ○ Treasury
Risk Management ○ Functions of
the Treasury Department
- Summary
- Self-Assessment
- Questions/Exercises
- Solutions/Answers

Unit 2: Receivables Management

- Introduction
- Objectives
- Terms of Payment
- Credit Policy Variables
- Credit Evaluation
- Monitoring Receivables
- Factoring
- Non-Performing Assets
- Summary
- Self-Assessment Questions
- Solutions/Answers

Unit 3: Inventory Management

- Introduction
- Objectives
- Reasons for Holding Inventory

Analysis:

Types of Data

Sampling

Descriptive – Summaries without

Management

- Techniques of Inventory Control ○
Traditional Techniques ○ Modern
Techniques
- Summary
- Self-Assessment
- Questions/Exercises
- Solutions/Answers

interpretation

Exploratory – No guarantee if

discoveries will hold in a new

sample Inferential, Causal

Predictive

Common Mistakes – Correlation

is not causation, Simpson's

paradox, Data

Dredging

Applications of Data Science

Data Science Life cycle

Unit 2: Portability and Statistics for Data Science

Statistics: Correlation

Probability: Dependence and

Independence, Conditional

Probability, Bayes Theorem,

Random Variables, Some basic

Distributions, the Normal

Distribution, The Central Limit

Theorem

Hypothesis: Statistical

Hypothesis

Testing, Confidence Intervals,

Unit 3: Data Preparation for

Analysis Data Preprocessing

Selection and Data Extraction

Data cleaning

Data Curation

Data Integration

Knowledge Discovery

Unit 4: Data Visualization and

Interpretation Different types of plots

Histograms

Boxplots

Scatter plots

Plots related to regression

Data Interpretation using

Examples

Block 2: Big Data and its

Management Unit 5: Big

Architecture Big Data and

Characteristics and Applications (Big Data and its importance, Four Vs) Big

data Application Structured vs semi-

structured and unstructured data Big

Data vs data warehouse

Distributed file system

Map Reduce and HDFS

Apache Hadoop 1 and 2

(YARN)

Hadoop Ecosystem – Name

node, data node, Job tracker

Unit 6: Programming using MapReduce

Map Reduce Operations

Loading data into HDFS

Executing the Map phase

Shuffling and sorting

Reduce phase execution.

Algorithms using map reduce –

Word counting, Matrix-Vector

Multiplication

Unit 7: Other Big data Architecture and

Tools

Apache SPARK framework HIVE

HBASE

Other tools

Unit 8: No SQL database

Column based

Graph based

Key-value pair based

Document based

Block 3: Big Data Analysis

Unit 9: Mining Big Data

Finding Similar Items

Jaccard Similarity of Sets

Similarity of Documents

Collaborative Filtering as a

Similar-Sets Problem Documents and Shingles

Distance Measures

Euclidean Distances

Jaccard Distance

Cosine Distance Edit Distance

Hamming Distance

Introduction to Other Techniques

Supervised Learning

Unsupervised Learning

Unit 10: Mining Data Streams

Model for Data Stream Processing

Data Stream Management Example

Queries of Data stream

Issues and challenges

Data sampling in data streams

Example of representation

sample

Filtering of data streams

Bloom filter

Algorithm to count different elements in stream

Unit 11: Link Analysis

Purpose of Link analysis

Page Ranking Different

mechanisms of finding

page Rank and their

problem

Web structure and

associated issues

Use of page rank in

search engines Page

Rank computation using

Mapreduce

Topic sensitive Page Rank

Link Spam

Hubs and Authorities

Unit 12: Web and Social Network Analysis

Introduction to Web Analytics

Advertising on the Web

Issues in On-Line Advertising

Advertising Opportunities on

Web

Direct Placement of Ads and its issues
 On-Line and Off-Line Algorithms
 Recommendation Systems
 Recommendation Systems Model and its Applications, The Utility Matrix
 The Long Tail
 Content-Based Recommendations
 Mining Social-Network
 Social Networks as Graphs
 Varieties of Social Networks
 Distance measure of social network
 Graphs
 Use of Clustering for social media

Block 4: Programming for Data Analysis

Unit 13: Basic of R Programming

Environment of R
 Data types, Variables, Operators, Factors
 Decision Making, Loops, Functions
 Data Structures in R
 Strings, Vector
 Lists, Frames
 Matrices, Arrays

MCS-227 Cloud Computing and IoT (4 Credits)

Unit 14: Data Interfacing and Visualisation in R

CSV, Excel files
 Binary files
 XML files JSON interface
 Database
 Web Data
 Data cleaning, Processing
 Bar Charts
 Box Plots Histograms Line Charts Scatter plots etc.

Unit 15: Data Analysis and R

Chi-square test
 Linear Regression
 Multiple Regression
 Logistic Regression
 Time Series Analysis

Unit 16: Advance Analysis using R

Decision Trees
 Random Forest
 Classification
 Clustering
 Association rules

After completing this course, the student will be able to:

- Understand the differences between the traditional computing and cloud computing
- Compare and contrast various deployment models and service delivery models of a cloud computing architecture.
- Understand the ways of virtualization
- Interpret the resource pooling, sharing and provisioning
- Understand the concept of scaling and load balancing in cloud
- Elaborate the need of security in cloud computing
- Define IoT and related terminology, technology and its applications
- Interpret the impact and challenges posed by IoT networks leading to new architectural models.
- Compare and contrast the deployment of smart objects and the technologies to connect them to network.
- Appraise the role of IoT protocols for efficient network communication.
- Elaborate the need Security in IoT.

- Illustrate different Case Studies from various sectors.

The topics includes in the course are:

Introduction to Cloud Computing, Traditional Computing Approaches, Comparison of Cluster, grid and Cloud Computing, Evolution of Cloud Computing, Benefits and Challenges, Cloud Deployment Models (Public, Private, Community and Hybrid), Service Delivery Models (IaaS, PaaS, SaaS etc..), Cloud Architecture, Resource Virtualization, Resource Pooling, Sharing and Provisioning, Scaling in the Cloud, Load Balancing, Security Issues in Cloud Computing.

IoT-An Introduction, Characteristics, IoT categories, Baseline Technologies of IoT, M2M and IoT, Multihoming, IoT Identification and Data protocols (IPv4, IPv6, MQTT, CoAP, SMPP, AMQP), Connectivity Technologies (IEEE, 802.15.4, Zigbee, 6LoWPAN, RFID, NFC, Bluetooth, Z-wave), IoT Application Development, Framework for IoT Applications, Implementation of Device Integration, Data Acquisition and Integration, Device Data Storage, Fog Computing, Edge Computing, Comparison of Cloud, Fog and Edge, IoT Case Studies (Smart Homes, Smart Grids, Smart Cities, Connected Vehicles, Industrial IoT.

Course Structure*

BLOCK 1: CLOUD COMPUTING FUNDAMENTALS AND VIRTUALIZATION

Unit 1: Cloud Computing: An Introduction

- Traditional Computing Approaches
- Evolution of Cloud Computing
- Comparison between Cluster, Grid and Cloud Computing
- Utility Computing
- Characteristics of Cloud Computing
- Benefits of Cloud Computing
- Applications of Cloud Computing
- Challenges of Cloud Computing

Unit 2: Cloud Deployment Models, Service Models and Cloud Architecture

- Cloud Deployment Models ○ Public Cloud ○ Private Cloud ○ Community Cloud ○ Hybrid Cloud
- Choosing Appropriate Deployment Model
- Service Delivery Models ○ Infrastructure As a Service (IaaS) ○ Platform As a Service (PaaS) ○ Software As a Service (SaaS) ○ Other Services (Security Management, Identity Management, Storage, Database,

Back-up, Compliance etc.)

- Cloud architecture
- Layers and Anatomy of the Cloud
- Network Connectivity in Cloud

Computing Unit 3: Resource Virtualization

- Virtualization and Underlying Abstraction ○ Virtualizing Physical Computing Resources
- Advantages of Virtualization
- Machine or Server Level Virtualization ○ Hosted Approach ○ Bare Metal Approach
- Exploring Hypervisor or V3irtual Machine Monitor ○ Hypervisor Based Virtualization Approaches (Full Virtualization, Para Virtualization, Hardware Assisted Virtualization)
- Operating System Level Virtualization
- Other Virtualizations (Network, Storage, Desktop)
- Xen Server Vs VM ware (comparison w.r.t features like Guest O/S support, Open Source, Live VM Snapshots for

Backups, Thin Provisioning, Asset Management and Configuration mapping, Dynamic Resource Allocation and Failover, Bare Metal Hypervisor, Graphics Support and Pricing, Licensing, Host Server Management, Storage Specifications etc.)

BLOCK 2: RESOURCE PROVISIONING, LOAD BALANCING AND SECURITY

Unit 4: Resource Pooling, Sharing and Provisioning

- Resource Pooling
- Resource Pooling Architecture ○ Computer Vs Server Pool ○ Storage Pool ○ Network Pool
- Resource Sharing ○ Multi Tenancy ○ Types of Tenancy ○ Tenancy at Different Level of Cloud Services
- Resource Provisioning and Approaches ○ Static Approach ○ Dynamic Approach ○ Hybrid Approach
- VM Sizing

Unit 5: Scaling

- Scaling primitives
- Scaling Strategies ○ Proactive Scaling ○ Reactive Scaling ○ Combinational Scaling
- Auto Scaling in Cloud
- Types of Scaling ○ Vertical Scaling or Scaling Up ○ Horizontal Scaling or Scaling Out

Unit 6: Load Balancing

- Importance of Load Balancing
- Goals of Load Balancing
- What are to Load Balance and how it is done
- Levels of Load Balancing ○ VM Provisioning ○ Resource Provisioning
- Categories of Load Balancing ○ Static Approach ○ Dynamic Approach
- Dynamic Load Balancing

Unit 7: Security Issues in Cloud Computing

- Threats to Cloud Security

- Infrastructure Security
- Information Security
- Identity Management and Access Control
- Cloud Security Design Principles • Security as a Service

BLOCK 3: IoT FUNDAMENTALS AND CONNECTIVITY TECHNOLOGIES

Unit 8: Internet of Things: An Introduction

- Introduction to IoT
- Characteristics of IoT
- IoT Categories
- IoT Enablers and Connectivity Layers
- Baseline Technologies of IoT
- Sensors ○ Characteristics of a Sensor ○ Classification of Sensors
- Actuators ○ Types of Actuators
- Computing Components (Arduino, Raspberry Pi),
- IoT Architecture
- Applications of IoT
- Challenges of IoT

Unit 9: IoT Networking and Connectivity Technologies

- M2M and IoT Technology
- Components of Networking
- Gateway Prefix Allotment
- Impact of Mobility on Addressing
- Multihoming
- IoT Identification and Data Protocols ○ (IPV4, IPV6, MQTT, CoAP, XMPP, AMQP)
- Connectivity Technologies ○ (IEEE 802.15.4, ZigBee, 6LoWPAN, RFID, NFC, Bluetooth, Z-wave etc.)

BLOCK 4: Application Development, Fog Computing and Case Studies

Unit 10: IoT Application Development

- Framework for IoT Applications
- Implementation of Device Integration
- Data Acquisition and Integration
- Device Data Storage

- Unstructured Data Storage on Cloud/Local Server
- Authentication, Authorization of Devices
- Security Aspects in IoT

Unit 11: Fog Computing and Edge Computing

- Introduction to Fog Computing
- Cloud Computing Vs Fog Computing
- Fog Architecture
- Working of Fog
- Advantages of Fog
- Applications of Fog
- Challenges in Fog
- Edge Computing
- Working of Edge Computing
- Cloud Vs Fog Vs Edge Computing
(w.r.t location of data processing, processing power and storage capabilities, purpose)
- Applications of Edge Computing

Unit 12: IoT Case Studies

- Smart Homes
- Smart Grids
- Smart Cities
- Connected Vehicles
- Industrial IoT

MCSL-228 AI and Machine Learning Lab (Credits 2)

Main objective of this laboratory course is to provide hands on exercises to the learners based on Artificial Intelligence and Machine Learning Course.

Lab Sessions:

- There will be 20 practical sessions (3 hours each) of which 10 sessions will be on AI and 10 sessions will be on machine learning.
- The practice problems for all 20 sessions will be listed session-wise in the lab manual.

MCSL-229 Cloud and Data Science Lab (Credits 2)

Main objective of this laboratory course is to provide hands on exercises to the learners based on Cloud Computing and Data Science Courses.

Lab Sessions:

- There will be 20 practical sessions (3 hours each) of which 10 sessions will be on cloud computing and 10 sessions will be on Data Science.
- The practice problems for all 20 sessions will be listed session-wise in the lab manual.

SEMESTER – IV

MCS-230 Digital Image Processing and Computer Vision (CREDITS - 4)

The course relates to the formation of fundamental understanding of the various concepts of Digital Image processing and Computer Vision. The content coverage will help the learners to get the insight of the subject both theoretically and practically.

Course Structure*

Block-1 Digital images Processing -I

Unit-1 Introduction to digital image – Digital image, Image acquisition, Digitization of images (Sampling and Quantization), Types of images, Image Characteristics (Brightness, luminance, contrast, intensity), Image resolution

Unit-2 Image Transformation - Definition of 1-D and 2-D signals, Orthogonal and Unitary transforms of 2-D signals,

Properties of Unitary Transforms

Unit-3 Image enhancement in spatial domain - Point operations, Contrast stretching, Clipping and thresholding,

Digital Negative, Intensity levels slicing, Bit - Basics of filtering in frequency domain, Image smoothing, Image sharpening, Image degradation model, Noise models (additive, Gaussian, Rayleigh, uniform, gamma, impulse), Inverse filtering, Wiener filtering.

Unit-7 Color image processing - Human Vision system, Color models (RGB, HIS, CMY)

Block-3 Computer Vision-I

Unit-9 Introduction to computer Vision, camera models, Transformations: Orthogonal, Euclidean, Affine and Projective

Unit-10: Single Camera: Camera Models, Perspective projection, Homography, Camera Calibration, Affine motion models

Unit-11 Multiple Cameras: Stereo Vision, Point correspondence, Epipolar geometry, Motion, Optical flow.

extraction.

Unit-4 Image Filtering Operations in spatial domain - Spatial averaging, Spatial low pass filtering, Spatial high pass filtering,

Median filtering, Min, Max filtering, Histogram modeling: Histogram equalization, Histogram specification.

Block-2 Digital images Processing –II

Unit-5 Transformation Techniques -

Transformations in the Frequency domain (DFT, DCT, DWT, Haar), Discrete Fourier Transform, Discrete Cosine Transform, Discrete Wavelet Transform, Haar

Transform

Unit-6 Image enhancement and Filtering

Block-4 Computer Vision-II

Unit-12 Object detection- Line detection, Region detection, Boundary detection, feature extraction techniques, image segmentation techniques

Unit-13 Object Recognition using Supervised Learning Approaches -

Supervised learning, Discriminant function (linear and nonlinear), Bayesian classification, Minimum distance classifiers.

Unit-14 Object Classification using Unsupervised Learning Approaches

- Unsupervised learning, Hierarchical Clustering, Partition based clustering, K-NN clustering.

The following are the objectives of this course:

- Introduce Mobile Communications
- Introduce Mobile Computing Architecture
- Overview of Pervasive Computing
- Introduce GSM and GPRS
- Introduce 4G and 5G Networks
- Discuss Database Management Issues in Mobile Computing
- Introduce Mobile Adhoc Networks
- Introduce WLAN and PAN protocols
- Introduce Virtual and Cloud Networks
- Introduce Mobile Internet Applications
- Introduce Mobile Application Languages
- Introduce Mobile Operating Systems
- Introduce Mobile Software Development Environments

Course Structure***Block-1: Introduction to Mobile Computing****Unit-1: Introduction to Mobile Communications**

Introduction

Objectives
Mobile Communication
Multiplexing (TDMA, CDMA, FDMA)
GSM
GPRS and 2.5G
3G
4G –LTE
Summary
Further Readings

Unit-2: Introduction to Mobile Computing Architecture

Introduction

Objectives
Mobile IP, Cellular and WLAN
IEEE 802.11X Networks
AdHoc Networks

Mobile Computing Operating System

Client Server Computing using Mobile

Mobile Computing Architecture
Design considerations for Mobile Computing

Mobile Computing and the Apps
Summary

Further Readings

Unit-3: Mobile Client Devices and Pervasive Computing

Introduction

Objectives
Smart Sensors, Actuators and Mobile Robotic Systems
Smart Home and Appliances
Automotive Systems
Limitations and Devices Design Considerations
Summary
Further Readings

Unit-4: GSM and GPRS

Introduction
Objectives
GSM Architecture
Public Land Mobile Network
(PLMN) Interface
Call Handling
Handover

61

SMS
GPRS
High Speed Circuit Switched
Data
WLL Application
Summary
Further Readings

Block-2: Mobile IP and Issues in Mobile Computing

Unit-5: 4G and 5G Networks

Introduction
Objectives
High Speed Packet Access
MIMO in HSPA
LTE and WIMAX16E
Ultra-Wide Band and Broadband
Wireless Access
4G Networks: HS-OFDM, LTE
Advanced and WiMax 16M
Features of 5G Networks
Summary
Further Readings

Unit-6: Mobile IP Network Layer

Introduction
Objectives
Mobile IP
IP Header: Encapsulation and
Routes Optimization
Mobility Binding
Cellular IP
Mobile IP with IPv6
Voice over IP
IP Security
Summary
Further Readings

Unit-7: Mobile Transport Layer

Introduction
Objectives
UDP and TCP
Indirect TCP
Snooping TCP

Mobile TCP
Summary
Further Readings

Unit-8: Database Management Issues in Mobile Computing

Introduction
Mobile Device Database
Management
Mobile Device Data Store
Methods
Client Server Computing with
Adaptation for Mobile Computing
Adaptation Software for Mobile
Computing
Summary
Further Readings

Block-3: Introduction to various Network Technologies

Unit-9: Mobile Adhoc Networks

Introduction
Objectives
Introduction to MANETs
Routing and Classifications of
Routing Algorithms
QoS in MANETs
Security in MANETs
Summary
Further Readings

Unit-10: WLAN and PAN protocols

Introduction
Objectives
Introduction to WLANs
Introduction to WAP
Introduction to WML
Bluetooth

WiMax
ZigBee and WiFi
Summary
Further Readings

Handheld Device Markup
Language and WML
HTML 5
Summary
Further Readings

Unit-11: Virtual and Cloud Networks

Introduction
Objectives
Wireless Enterprise Networks
Virtual Networks
Mobile Cloud Networks
Summary
Further Readings

Unit-15: Mobile Application Languages

Introduction
Objectives
Introduction to J2EE

Unit-12: Mobility, Portability, Replication and Clustering

Introduction
Objectives
Mobile Data Management
Data Replication Schemes
Adaptive Clustering
Summary
Further Readings

Block-4: Introduction to Mobile Software Environments

Unit-13: Smart Client and Enterprise Server based Architecture

Introduction
Objectives
Introduction to Smart Client
Architecture
Data Synchronization Formats
Data Synchronization at Clients
and Servers
Mobile Devices Support
Infrastructure and Management
Summary
Further Readings

Unit-14: Mobile Internet Applications

Introduction
Objectives
Introduction to Mobile
Applications Development
Introduction to XML

Introduction to J2ME
Introduction to Python
Summary
Further Readings

Unit-16: Mobile Operating Systems and Development Environments

Introduction
Objectives

Introduction to Mobile Operating Systems
Application Programming
Linux for Mobile Devices
Development Process
Development Tools and Emulators
Apple IOS
Android
Summary
Further Readings

*** The course structure may be subject to changes.**

MCSP-232 PROJECT

(CREDITS-12)

The objective of the MCA project work is to develop quality software solution by following the software engineering principles and practices. It is only possible when a learner goes about with the task independently. During the development of the project the students should involve in all the stages of the software development life cycle like requirements engineering, systems analysis, systems design, software development, testing strategies and documentation with an overall emphasis on the development of reliable software systems. The primary emphasis of the project work is to understand and gain the knowledge of the principles of software engineering practices, so as to participate and manage a large software engineering projects in future.

Students are encouraged to spend efforts equivalent to 12 credits working on a project preferably in a software industry or any research organization. Topics selected should be complex and large enough to justify as a MCA project. The courses studied by the students during the MCA programme provide them the comprehensive background to work on diverse application domains. Students should strictly follow and adhere to the project guidelines. Project Guidelines will be prepared and uploaded on to the IGNOU website/printed.

5. EVALUATION SCHEME

Completion of the programme requires successful completion of both assignment component and the Term-end Examination component for each course in the programme. The total numbers of courses in this MCA programme are 22 (including a Project course) and the total number of credits is 80. In addition, two bridge courses, MCS-201 and MCS208 are required to be successfully completed by those *students who are required to register for these courses based on eligibility conditions for admission to Master of Computer Applications programme.*

Evaluation for each course of MCA programme (except project course) and bridge courses covers two aspects:

- Continuous evaluation through Assignment with a weightage of 30% (please refer to the table below). *Viva- voce is compulsory for all the Assignments for which 20 marks are allocated.*
- Term-end examination with a weightage of 70% (please refer to the table below).